Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Systems of Linear Inequalities by Graphing

Name: ________________
Date: ____________

Advanced (Carolina Reaper)

Q1. What is the purpose of graphing multiple inequalities on the same coordinate plane?
(A) To find the intersection point of two lines
(B) To identify overlapping solution regions
(C) To determine the slope of a line
(D) To calculate the area of a triangle
(E) To simplify an algebraic expression

Q2. Given the inequality $x + 2y \geq 4$, which of the following is a possible solution?
(A) $(0, 0)$
(B) $(0, 2)$
(C) $(2, 0)$
(D) $(1, 1)$
(E) $(-1, -1)$

Q3. What is the term for the area where the solution regions of multiple inequalities overlap?
(A) Feasible region
(B) Solution set
(C) Graph
(D) Coordinate plane
(E) Inequality

Q4. How many solution regions can be formed by graphing two linear inequalities?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q5. Given the system of inequalities $x + y \leq 2$ and $x - y \geq -1$, which of the following points is in the feasible region?
(A) $(0, 0)$
(B) $(1, 1)$
(C) $(2, 0)$
(D) $(0, 2)$
(E) $(-1, 0)$

Q6. What is the equation of the boundary line for the inequality $2x - 3y > 5$?
(A) $2x - 3y = 5$
(B) $2x + 3y = 5$
(C) $x - 2y = 3$
(D) $x + 2y = 3$
(E) $3x - 2y = 5$

Q7. Which of the following statements is true about the graph of the inequality $y \geq 2x - 1$?
(A) The line $y = 2x-1$ is the boundary line and the shaded region is above the line
(B) The line $y = 2x-1$ is the boundary line and the shaded region is below the line
(C) The line $y = 2x+1$ is the boundary line and the shaded region is above the line
(D) The line $y = 2x+1$ is the boundary line and the shaded region is below the line
(E) The line $y = -2x + 1$ is the boundary line and the shaded region is above the line

Q8. Given the system of inequalities $x + y \leq 3$ and $x - y \geq -2$, which of the following is the correct graph of the feasible region?
(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D
(E) Graph E

Open Ended Questions (FRQ)

Q9. Graph the system of inequalities $x + 2y \geq 4$ and $x - y \leq 1$. Shade the feasible region and label all boundary lines.

Q10. A company produces two products, A and B, using two machines, X and Y. Machine X can produce 2 units of product A and 1 unit of product B per hour. Machine Y can produce 1 unit of product A and 3 units of product B per hour. The company has 8 hours of machine time per day on machine X and 12 hours of machine time per day on machine Y. The company wants to produce at least 16 units of product A and at least 12 units of product B per day. Set up a system of linear inequalities to represent this situation and graph the feasible region.

Chef's Tips

- Tip 1: Make sure students understand the concept of a feasible region and how to graph it.
- Tip 2: Emphasize the importance of labeling all boundary lines and shading the feasible region correctly.
- Tip 3: Encourage students to use different colors to distinguish between different inequalities when graphing.